

11.15 Do Now: Introduction to Periodic Functions (no calculator)

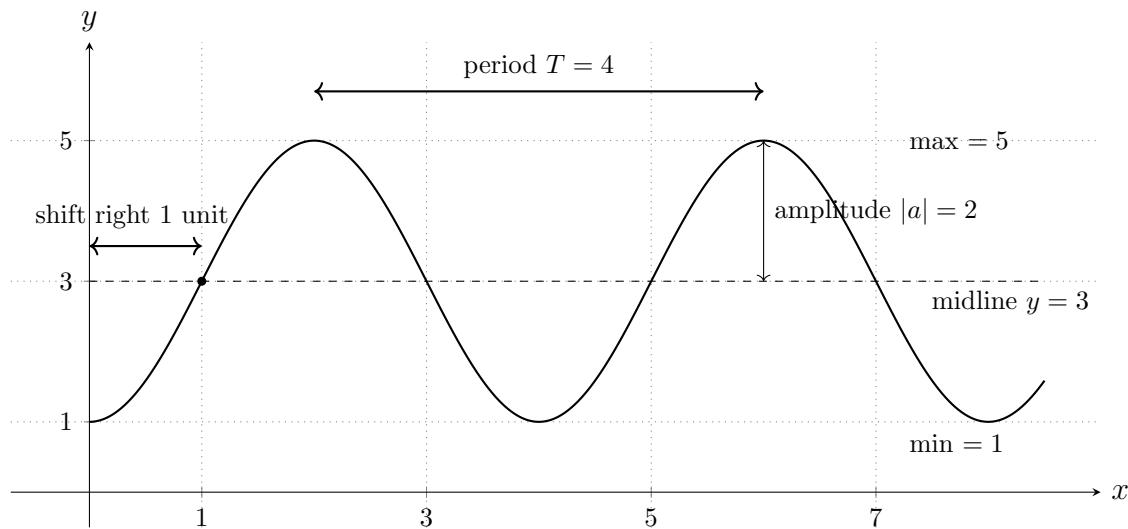
The graphs of $\sin$ and $\cos$ can be stretched, shifted, and translated to model periodic phenomena (tides, springs, rotating wheels). The general form is

$$f(x) = a \sin(b(x + c)) + d \quad \text{or} \quad f(x) = a \cos(b(x + c)) + d,$$

where each of a , b , c , d controls one feature of the graph:

$ a $	amplitude	half of $(\max - \min)$; vertical stretch
b	frequency factor	period $T = \frac{2\pi}{b}$
c	phase shift	horizontal translation by $-c$
d	vertical shift	midline $y = d$

Example. $f(x) = 2 \sin\left(\frac{\pi}{2}(x - 1)\right) + 3$ has $a = 2$, $b = \frac{\pi}{2}$, $c = -1$, $d = 3$, period $T = 4$.



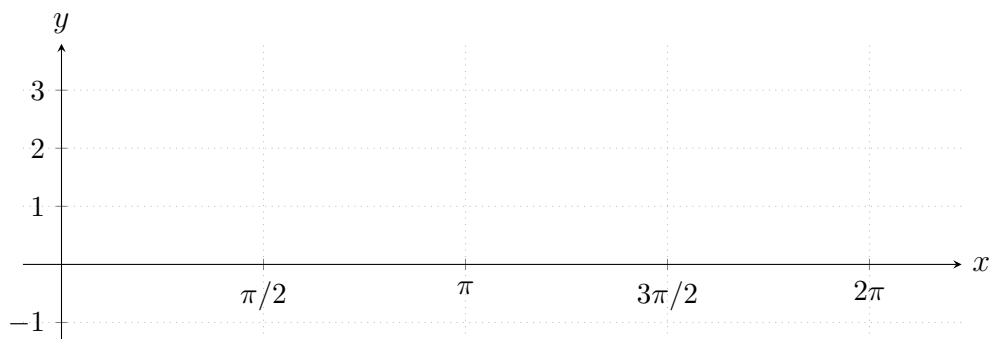
Reading a , b , c , d from a graph (inverse problem).

- $a = \frac{\max - \min}{2}$ (amplitude); $d = \frac{\max + \min}{2}$ (midline)
- $b = \frac{2\pi}{T}$, where T is the period (peak-to-peak distance)
- c from a known point on the graph — often a maximum, minimum, or zero crossing

Notation note. Some IB problems write $f(x) = a \sin(b(x - c)) + d$ with a minus sign. The amplitude, period, and midline are unchanged; only the sign of c flips.

Practice.

1. For the function $f(x) = 3 \sin(2x) - 1$, write down:
 - (a) the amplitude;
 - (b) the period;
 - (c) the equation of the midline;
 - (d) the maximum value of f .
2. For the function $g(x) = 4 \cos\left(\frac{\pi}{6}(x - 2)\right) + 5$, find:
 - (a) the amplitude;
 - (b) the period;
 - (c) the equation of the midline;
 - (d) the smallest value of $x \geq 0$ at which g is a maximum.
3. Sketch $y = 2 \sin x + 1$ for $0 \leq x \leq 2\pi$. Mark the maximum value, minimum value, and the y -intercept.



4. A function f has the form $f(x) = a \sin(b(x + c)) + d$, with $a > 0$ and $b > 0$. The function has:
 - maximum value 7 and minimum value 1;
 - period 8;
 - first maximum at $x = 1$.

Find the value of a , b , c , and d .